The University of Azad Jammu & Kashmir, Muzaffarabad



Student Name:	Adeela Bashir
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Instructor:	Engr.Mam.Asma

Department of Software Engineering

Q1: Activity Diagram of FoodPanda Application Using Swimlanes

Objective

To represent the process of ordering food through the FoodPanda application using an activity diagram with swimlanes.

Introduction

FoodPanda is an online food ordering application that allows customers to browse restaurants, select food items, place orders, and receive food at their location. The activity diagram illustrates the sequence of activities performed by the customer, FoodPanda system, restaurant, and delivery partner.

Swimlanes

1. Customer (User)
2. FoodPanda System
3. Restaurant
4. Delivery Partner

Main Activities

1. Customer opens the FoodPanda application.
2. Customer browses available restaurants.
3. Customer selects food items and places an order.
4. The system displays order details and payment options.
5. The system processes the payment.
6. The restaurant receives the order and prepares the food.
7. The restaurant marks the order as ready.
8. The delivery partner picks up the order.
9. The delivery partner delivers the food to the customer.
10. The customer receives the order and may provide feedback.

Conclusion

The activity diagram illustrates the sequence of activities involved in online food ordering and shows the responsibilities of the customer, FoodPanda system, restaurant, and delivery partner.

Activity Diagram

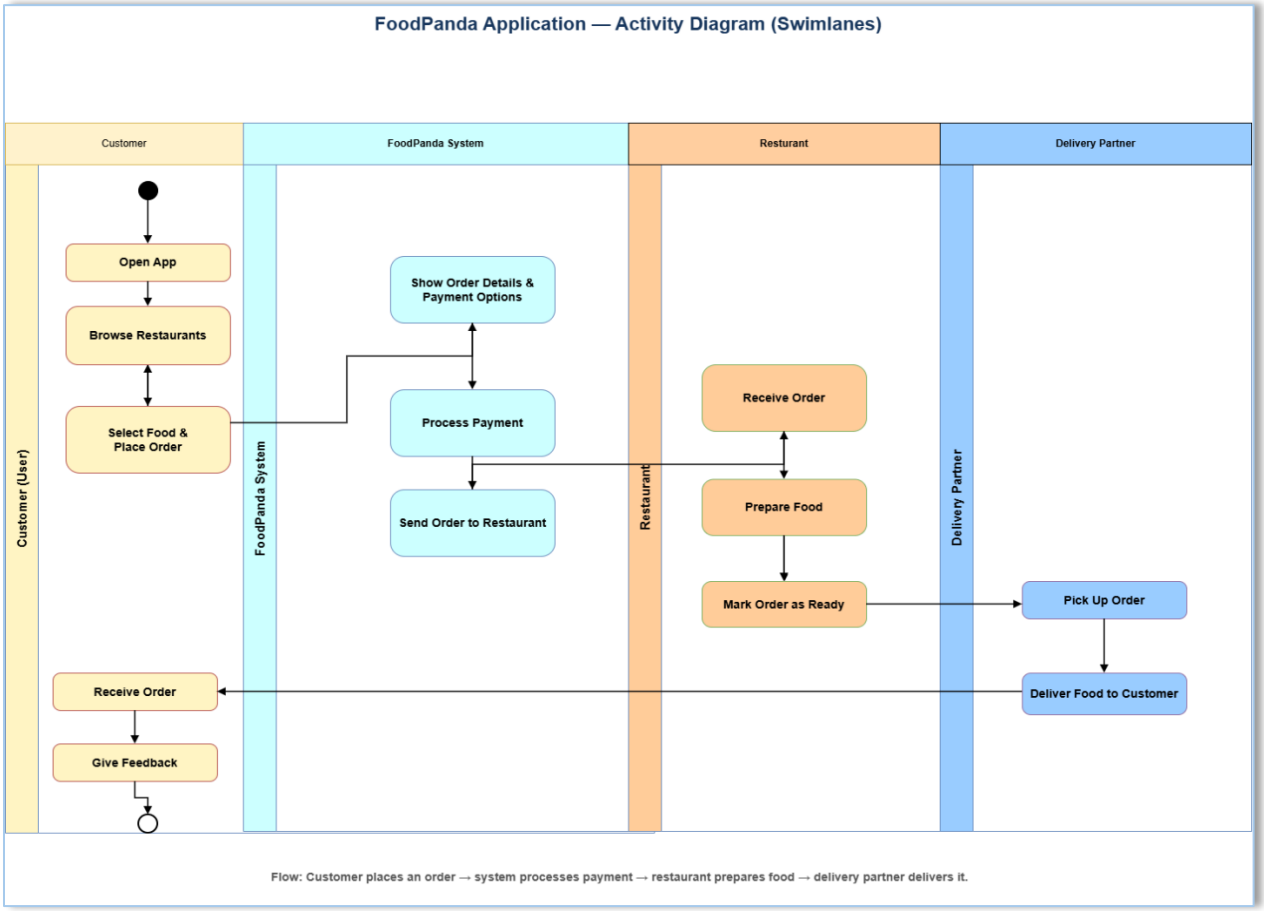


Fig: Activity Diagram of FoodPanda Application Using Swimlanes

Q2: State Diagram for a Toaster

Objective

To represent the different states of a toaster and identify its activities, triggered events, and triggerless events.

States and Activities

1. Idle
The toaster waits for the user to start the operation. The heating elements are off.

2. Heating
The heating elements are switched on, and the toaster begins heating.

3. Toasting
The timer runs while the bread is being toasted.

4. Done
The toasting process is complete, and the heating elements are switched off.

Triggered Events and Actions

1. Push lever: Starts heating and changes the state from Idle to Heating.
2. Cancel: Stops heating and returns the toaster to Idle.
3. Timer finished: Stops the toasting process and moves the toaster to Done.

Triggerless Events and Actions

1. After the heating period: The toaster moves from Heating to Toasting automatically.
2. After the toasting duration: The toaster moves from Toasting to Done automatically.

Conclusion

The state diagram represents the operating cycle of a toaster, from waiting for the user to start the process until the toast is ready.

State Diagram

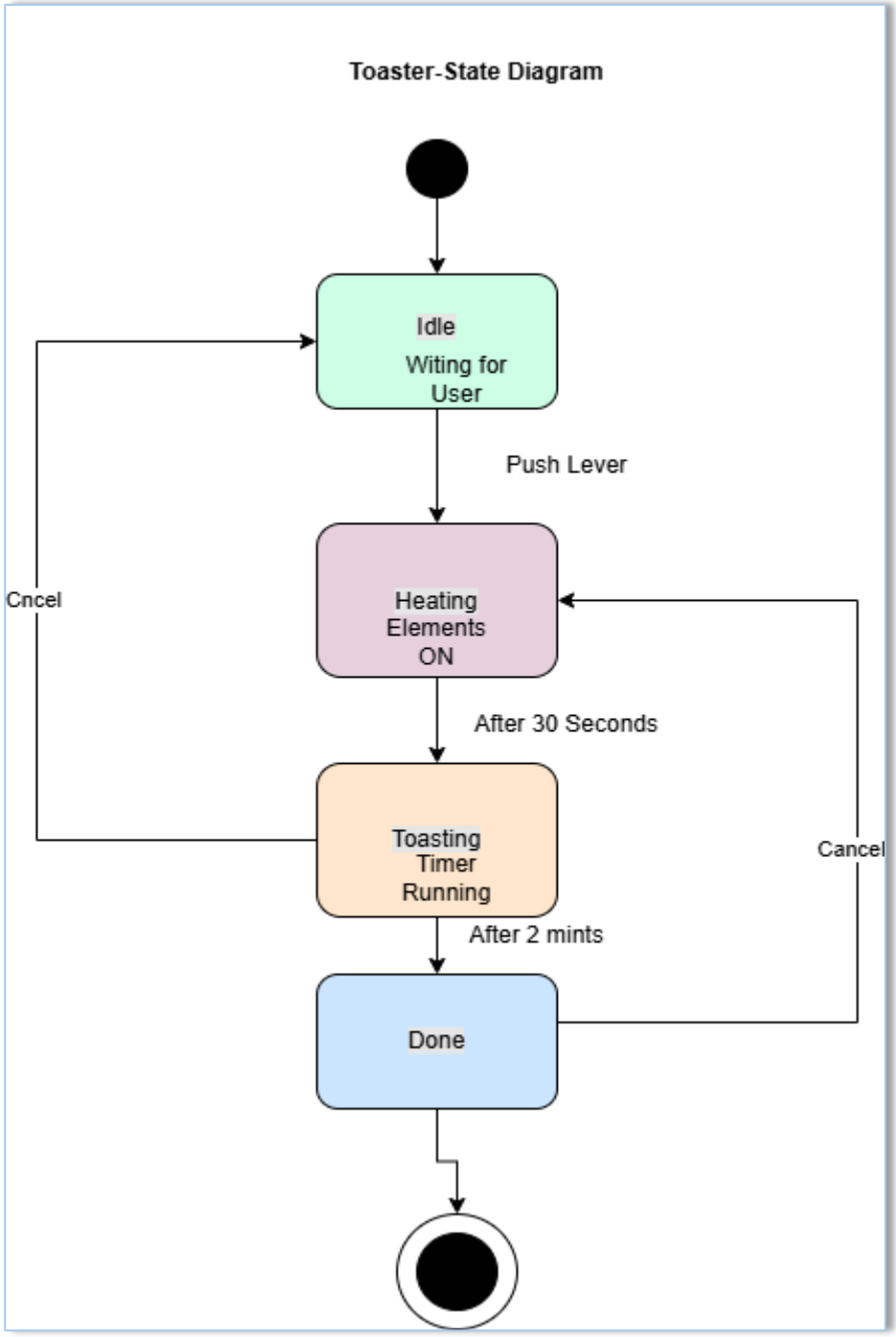


Fig: State diagram for toaster

Q3: State Machine Diagram of an Elevator/Lift

Objective

To represent the main states of an elevator and show the events that cause transitions between those states.

States and Activities

1. Idle
The elevator is stationary, and its doors are closed.

2. Moving Up
The elevator moves upward in response to an upward floor request.

3. Moving Down
The elevator moves downward in response to a downward floor request.

4. Door Open
The elevator stops at a requested floor and opens its doors to allow passengers to enter or exit.

Events and Transitions

1. Call Up: The elevator changes from Idle to Moving Up.
2. Call Down: The elevator changes from Idle to Moving Down.
3. Floor Reached: The elevator stops moving and changes to Door Open.
4. Door Close: The elevator returns to Idle when the doors close.
5. New Floor Request: The elevator may begin moving again in the requested direction.

Conclusion

The state machine diagram illustrates the main operating states of an elevator and the events that cause transitions between them.

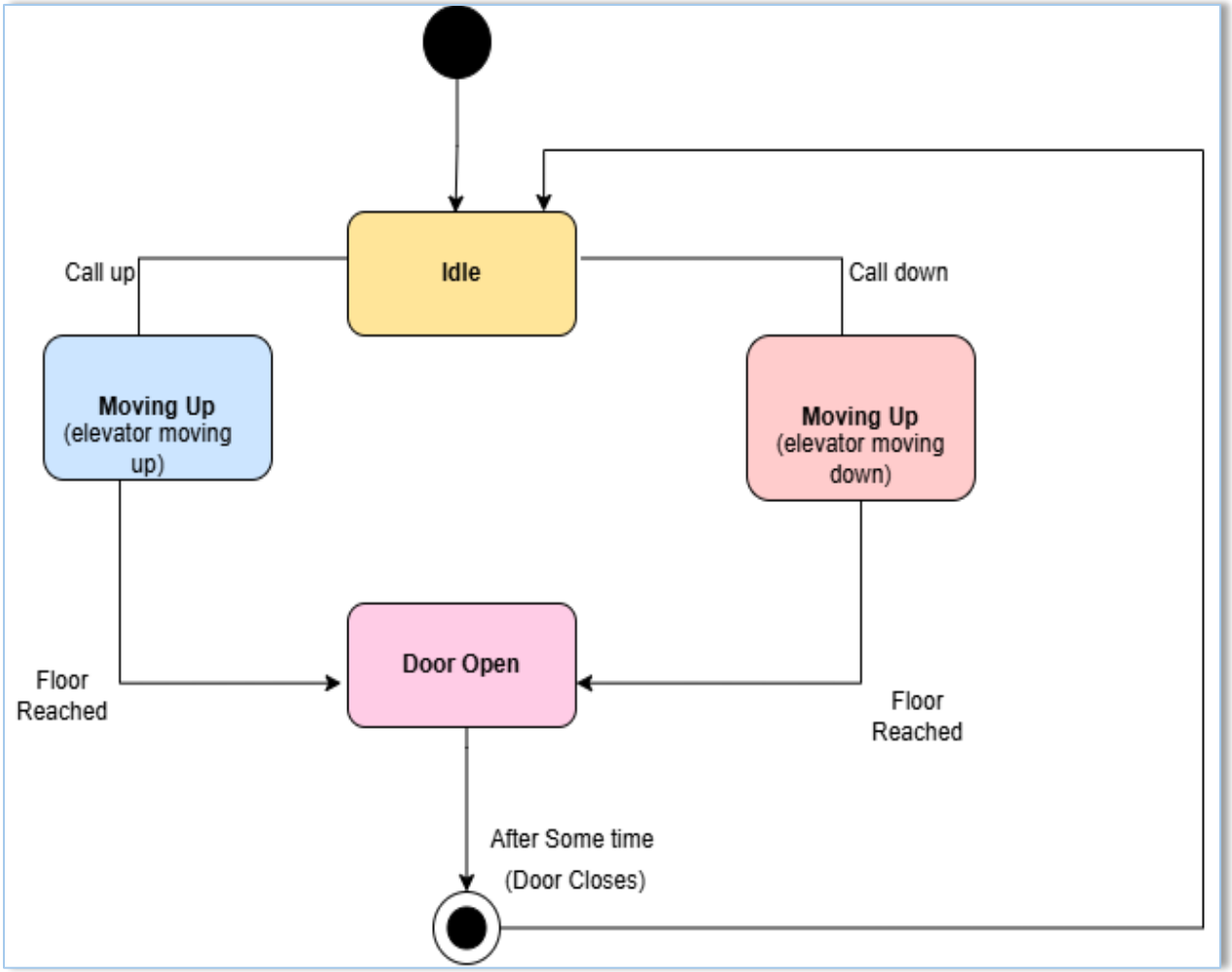


Fig: State machine diagram for elevator/ lift

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